

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 2

6031/2
1 hour 30 minutes

JUNE 2025 SESSION

Candidates answer on the question paper.

Additional materials:

Data Booklet

Electronic Calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question

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1	
2	
3	
4	
5	
6	
TOTAL	

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Answer all questions

- 1 (a) (i) Define a *Bronsted-Lowry base*.

- (ii) Ammonia can dissociate in water as shown in the equation;



Identify the conjugate acid-base pairs.

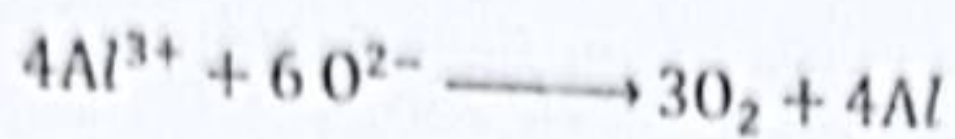
- (iii) Write an expression for the base dissociation constant of ammonia.

- (iv) Deduce the value of K_b of ammonia given that a $1 \times 10^{-2} \text{ mol dm}^{-3}$ ammonia solution has a pH of 11.12.

[6]



- (b) The overall equation for the electrolytic purification of aluminium is shown:



A current of 4.5 A was passed through the circuit for $2\frac{1}{2}$ hours.

Calculate the

- (i) quantity of charge used,
- (ii) number of moles of electrons used,
- (iii) number of moles of oxygen released,
- (iv) volume of oxygen produced at 25 °C.

[4]
[Total:10]



- 2 (a) (i) State any **one** advantage and any **one** disadvantage of fuel cells.

advantage _____

disadvantage _____

- (ii) Give the electrode reactions of a hydrogen-oxygen fuel cell using potassium hydroxide as an electrolyte.

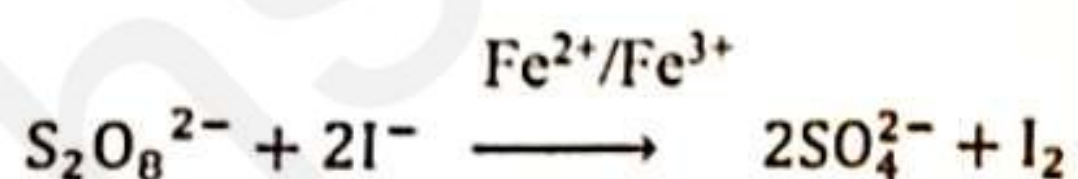
anode _____

cathode _____

- (iii) Calculate the E^0 cell of the hydrogen-oxygen fuel cell.

[5]

- (b) Peroxodisulphate ions react with iodide ions as follows:



- (i) Write chemical equations for the reactions to show the role of Fe^{2+}/Fe^{3+} catalyst.

- (ii) State, with a reason, the type of catalysis shown.



- (c) State the oxidation state of iodine in NaIO_3 .

[5]

- 3 (a) (i) Compare the reactions of barium and calcium with oxygen.

- (ii) Describe and explain the trend in ionisation energies of Group (II) elements down the group.

[5]

- (b) (i) Describe the structure, bonding and acid base nature of silicon dioxide.

structure _____

bonding _____

acid-base nature _____

- (ii) State any one use of ceramics based on their properties.

[5]

[Total:10]



4

The structure of an organic compound is shown in Fig. 4.1.

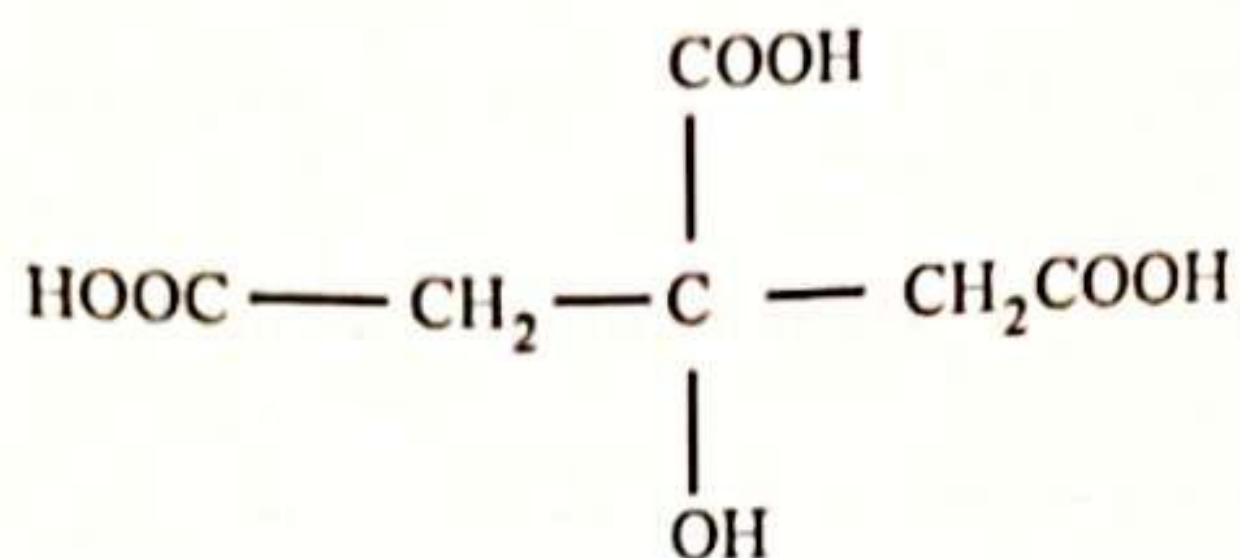


Fig. 4.1

- (a) (i) Give the molecular formula of the organic compound.
- _____
- (ii) Predict, with a reason, the solubility of the compound in water.
- _____
- _____
- _____
- (iii) Write the displayed formula of the product(s) formed when the compound reacts with
1. Na metal,
 2. $\text{NaOH}_{(\text{aq})}$.

[5]



- (b) (i) Give the structure of 2-aminoethanoic acid.

- (ii) Write an equation for the formation of a dipeptide from two aminoethanoic acid molecules.

- (iii) Draw the zwitterion formed from 2-aminoethanoic acid.

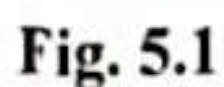
- (iv) Show the amphoteric nature of 2-aminoethanoic acid.

[5]

[Total:10]



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- (i) State conditions under which each of the compounds reacts with sodium hydroxide.
- S _____
- T _____
- (ii) Explain the differences on the ease of hydrolysis between S and T.
- _____
- _____
- _____
- (iii) Give the displayed formula of the product(s) formed when T reacts with ethylamine.

[4]



- (b) An organic compound, **P**, undergoes a reaction shown in Fig. 5.2.

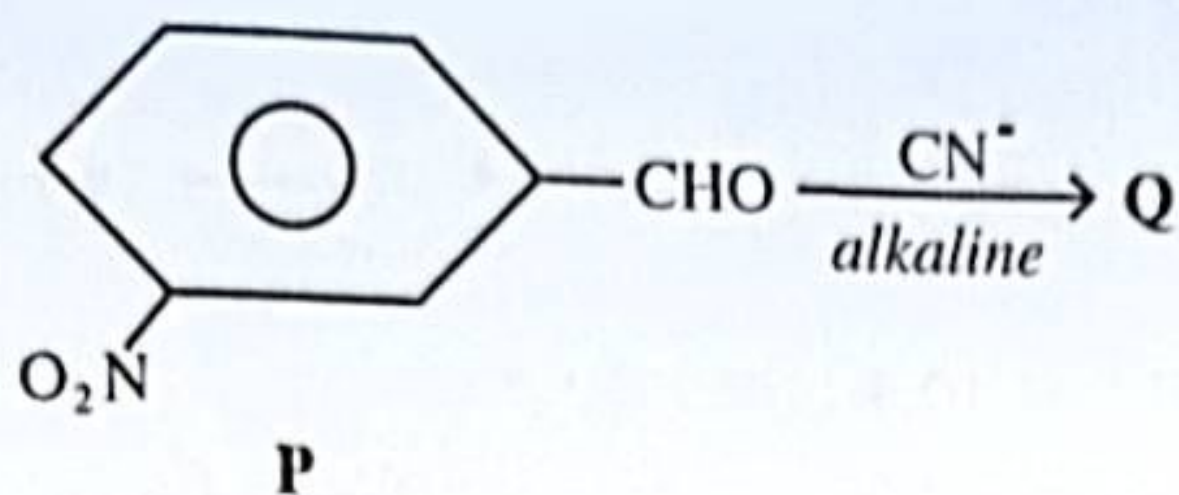


Fig. 5.2

- (i) Name compound **P**.

- (ii) Draw the structure of the organic product **Q**.

- (iii) Name the type of reaction shown in Fig 5.2.



(iv) Draw the isomers of Q.

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[4]
[Total: 10]

6 (a) A complex ion reacts as shown in the equation.



(i) Define the term *complex ion*.

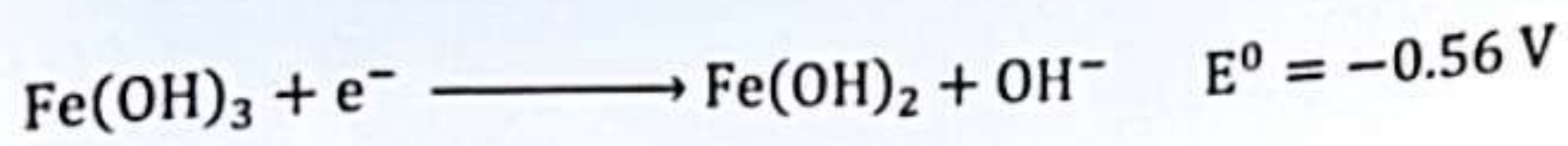
(ii) Write an expression for the stability constant, K_{stab} , for the formation of $[\text{Fe}(\text{CN})_6]^{3-}$.

(iii) Explain the significance of a K_{stab} value in the reaction.

[3]



- (b) Reduction potentials of $\text{Fe}^{3+}/\text{Fe}^{2+}$ system were measured in different solutions



Explain the observed differences in E° values.

_____ [2]

- (c) Ion exchange is one of the methods used to manage aqueous industrial waste.

- (i) State any one pollutant that can be found in aqueous industrial waste.

- (ii) Describe the ion exchange process.

- (iii) Suggest one limitation of ion exchange as a waste management method.

_____ [5]

[Total:10]

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